

FMUSER FUQ-4655K Encoder Modulator 12 HDMI Inputs 256 IP Inputs MPEG2 H.264 Encoding DVB-C-DVB-T-ATSC-ISDB-T Modulation Web Management CATV Headend Application *Product*

Overview

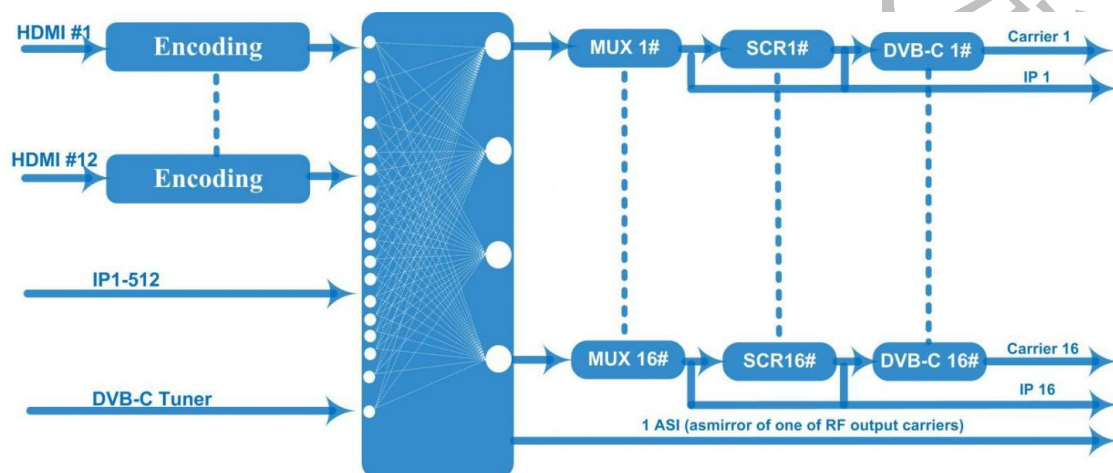


The FMUSER FUQ-4655L Encoder Modulator integrates encoding, multiplexing, scrambling, and modulating functions. It processes signals from 12 HDMI inputs and IP inputs, encoding to MPEG2 or H.264, and outputs via RF for DVB-C, DVB-T/ATSC, or ISDB-T standards. Managed through web-based interface for control and updates. Ideal for small CATV headend systems, broadcasting, and IP streaming applications.

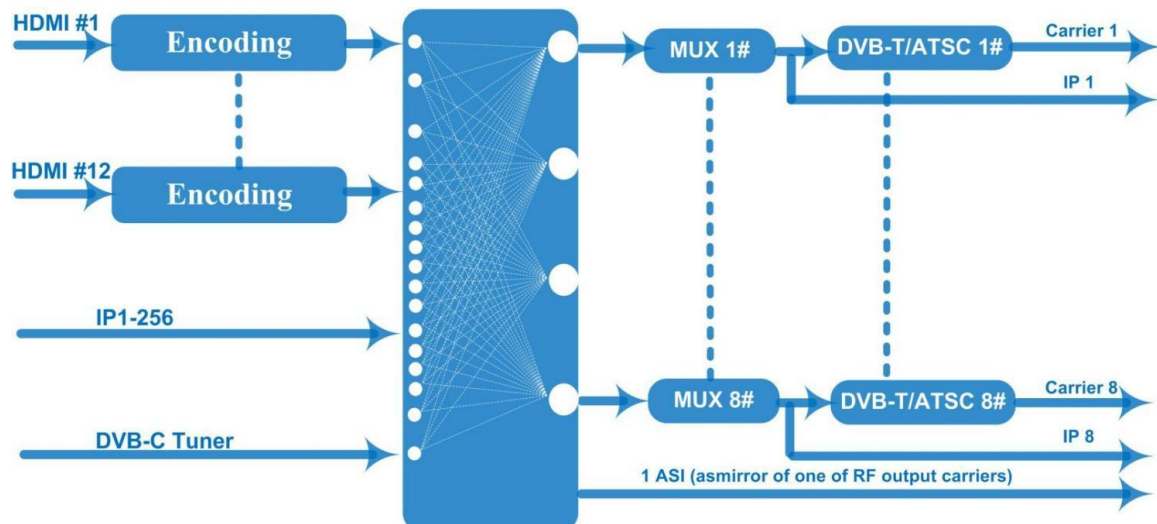
Key feature

Supports 12 HDMI inputs with MPEG2 & H.264 encoding, Features DVB-C, DVB-T/ATSC, and ISDB-T modulation for RF output, Integrated multiplexing and scrambling with PID remapping, Provides IP I/O over UDP/RTP/RTSP protocols, Web-based management for easy control

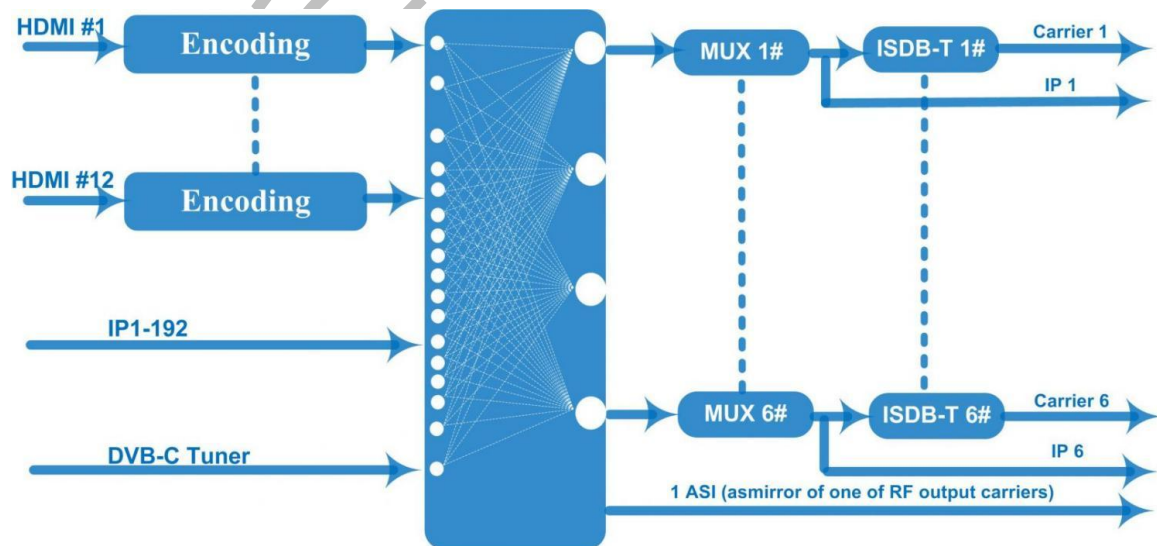
Principle Chart



DVB-C Principle Chart



DVB-T/ATSC Principle Chart



ISDB-T Principle Chart

Technical specification

Input	12 HDMI inputs 12 CC(closed caption) input, BNC interface 1 DVB-C(AT SC optional) Tuner for re-mux, F type interface 512 IP input over UDP and RTP, DATA1, RJ 45—DVB-C RF out 256 IP input over UDP and RTP, DATA1, RJ 45—DVB-T /ATSC RF out 192 IP input over UDP and RTP, DATA1, RJ 45—ISDB RF out	12 HDMI inputs 12 CC(closed caption) input, BNC interface 1 DVB-C(AT SC optional) Tuner for re-mux, F type interface 512 IP input over UDP and RTP, DATA1, RJ 45—DVB-C RF out 256 IP input over UDP and RTP, DATA1, RJ 45—DVB-T /ATSC RF out 192 IP input over UDP and RTP, DATA1, RJ 45—ISDB RF out	12 HDMI inputs 12 CC(closed caption) input, BNC interface 1 DVB-C(AT SC optional) Tuner for re-mux, F type interface 512 IP input over UDP and RTP, DATA1, RJ 45—DVB-C RF out 256 IP input over UDP and RTP, DATA1, RJ 45—DVB-T /ATSC RF out 192 IP input over UDP and RTP, DATA1, RJ 45—ISDB RF out	12 HDMI inputs 12 CC(closed caption) input, BNC interface 1 DVB-C(AT SC optional) Tuner for re-mux, F type interface 512 IP input over UDP and RTP, DATA1, RJ 45—DVB-C RF out 256 IP input over UDP and RTP, DATA1, RJ 45—DVB-T /ATSC RF out 192 IP input over UDP and RTP, DATA1, RJ 45—ISDB RF out	12 HDMI inputs 12 CC(closed caption) input, BNC interface 1 DVB-C(AT SC optional) Tuner for re-mux, F type interface 512 IP input over UDP and RTP, DATA1, RJ 45—DVB-C RF out 256 IP input over UDP and RTP, DATA1, RJ 45—DVB-T /ATSC RF out 192 IP input over UDP and RTP, DATA1, RJ 45—ISDB RF out	12 HDMI inputs 12 CC(closed caption) input, BNC interface 1 DVB-C(AT SC optional) Tuner for re-mux, F type interface 512 IP input over UDP and RTP, DATA1, RJ 45—DVB-C RF out 256 IP input over UDP and RTP, DATA1, RJ 45—DVB-T /ATSC RF out 192 IP input over UDP and RTP, DATA1, RJ 45—ISDB RF out
Video	Resolution	Input	Input	1920*1080_60p, 1920*1080_50p, 1920*1080_60i, 1920*1080_50i, 1280*720_60p, 1280*720_50P, 720*480_60i, 720*576_50i	1920*1080_60p, 1920*1080_50p, 1920*1080_60i, 1920*1080_50i, 1280*720_60p, 1280*720_50P, 720*480_60i, 720*576_50i	1920*1080_60p, 1920*1080_50p, 1920*1080_60i, 1920*1080_50i, 1280*720_60p, 1280*720_50P, 720*480_60i, 720*576_50i

Video	Resolution	Output	Output	1920*1080_60p, 1920*1080_50p, 1920*1080_60i, 1920*1080_50i, 1440*1080_60i, 1440*1080_50i, 1280*720_60p, 1280*720_50P, 720*576_50p, 720*576_50i, 720*576_30p, 720*576_25p, 720*480_60p, 720*480_60i, 720*480_30p, 720*480_25p, 320*240_60p, 320*240_50p, 320*240_30p, 320*240_25p, 320*180_60p, 320*180_50p, 320*180_30p, 320*180_25p, 960*540_50i, 704*576_50i, 704*480_60i, 640*576_50i, 640*480_60i, 544*768_50i, 544*480_60i,	1920*1080_60p, 1920*1080_50p, 1920*1080_60i, 1920*1080_50i, 1440*1080_60i, 1440*1080_50i, 1280*720_60p, 1280*720_50P, 720*576_50p, 720*576_50i, 720*576_30p, 720*576_25p, 720*480_60p, 720*480_60i, 720*480_30p, 720*480_25p, 320*240_60p, 320*240_50p, 320*240_30p, 320*240_25p, 320*180_60p, 320*180_50p, 320*180_30p, 320*180_25p, 960*540_50i, 704*576_50i, 704*480_60i, 640*576_50i, 640*480_60i, 544*768_50i, 544*480_60i,	1920*1080_60p, 1920*1080_50p, 1920*1080_60i, 1920*1080_50i, 1440*1080_60i, 1440*1080_50i, 1280*720_60p, 1280*720_50P, 720*576_50p, 720*576_50i, 720*576_30p, 720*576_25p, 720*480_60p, 720*480_60i, 720*480_30p, 720*480_25p, 320*240_60p, 320*240_50p, 320*240_30p, 320*240_25p, 320*180_60p, 320*180_50p, 320*180_30p, 320*180_25p, 960*540_50i, 704*576_50i, 704*480_60i, 640*576_50i, 640*480_60i, 544*768_50i, 544*480_60i,
Video	Encoding	MPEG2 & MPEG4 AVC/H.264	MPEG2 & MPEG4 AVC/H.264	MPEG2 & MPEG4 AVC/H.264	MPEG2 & MPEG4 AVC/H.264	MPEG2 & MPEG4 AVC/H.264

Video	Bit-rate	0.8~19Mbps for H.264 encoding 1~19Mbps for MPEG-2 encoding	0.8~19Mbps for H.264 encoding 1~19Mbps for MPEG-2 encoding	0.8~19Mbps for H.264 encoding 1~19Mbps for MPEG-2 encoding	0.8~19Mbps for H.264 encoding 1~19Mbps for MPEG-2 encoding	0.8~19Mbps for H.264 encoding 1~19Mbps for MPEG-2 encoding
Video	Rate Control	CBR/VBR	CBR/VBR	CBR/VBR	CBR/VBR	CBR/VBR
Video	GOP Structure	GOP B Frame: 0-3, GOP P Frame: 0-6	GOP B Frame: 0-3, GOP P Frame: 0-6	GOP B Frame: 0-3, GOP P Frame: 0-6	GOP B Frame: 0-3, GOP P Frame: 0-6	GOP B Frame: 0-3, GOP P Frame: 0-6
Video	Advanced Pretreatment	De-interlacing, noise reduction	De-interlacing, noise reduction	De-interlacing, noise reduction	De-interlacing, noise reduction	De-interlacing, noise reduction
Audio	Encoding	MPEG1 Layer II, MPEG2-AAC, MPEG4-AAC, Dolby Digital AC3 (2.0) encoding (Optional), AC3 (2.0/5.1) passthrough	MPEG1 Layer II, MPEG2-AAC, MPEG4-AAC, Dolby Digital AC3 (2.0) encoding (Optional), AC3 (2.0/5.1) passthrough	MPEG1 Layer II, MPEG2-AAC, MPEG4-AAC, Dolby Digital AC3 (2.0) encoding (Optional), AC3 (2.0/5.1) passthrough	MPEG1 Layer II, MPEG2-AAC, MPEG4-AAC, Dolby Digital AC3 (2.0) encoding (Optional), AC3 (2.0/5.1) passthrough	MPEG1 Layer II, MPEG2-AAC, MPEG4-AAC, Dolby Digital AC3 (2.0) encoding (Optional), AC3 (2.0/5.1) passthrough
Audio	Sampling rate	48KHz	48KHz	48KHz	48KHz	48KHz
Audio	Bit-rate	64Kbps-320kbps each channel	64Kbps-320kbps each channel	64Kbps-320kbps each channel	64Kbps-320kbps each channel	64Kbps-320kbps each channel
Multiplexing	Maximum PID Remapping	180 input per channel	180 input per channel	180 input per channel	180 input per channel	180 input per channel
Multiplexing	Function	PID remapping (automatically or manually)	PID remapping (automatically or manually)	PID remapping (automatically or manually)	PID remapping (automatically or manually)	PID remapping (automatically or manually)

Multiplexing	Function	Accurate PCR adjusting	Accurate PCR adjusting	Accurate PCR adjusting	Accurate PCR adjusting	Accurate PCR adjusting
Multiplexing	Function	Generate PSI/ SI table automatically	Generate PSI/ SI table automatically	Generate PSI/ SI table automatically	Generate PSI/ SI table automatically	Generate PSI/ SI table automatically
Scrambling	Maximum simulcrypt CA	4	4	4	4	4
Scrambling	Standard	ETR289, ETSI 101197, ETSI 103197	ETR289, ETSI 101197, ETSI 103197	ETR289, ETSI 101197, ETSI 103197	ETR289, ETSI 101197, ETSI 103197	ETR289, ETSI 101197, ETSI 103197
Scrambling	Connection	Local/remote connection	Local/remote connection	Local/remote connection	Local/remote connection	Local/remote connection

Modulation	DVB-C	QAM Channel: 16 non-adjacent carriers output (maximum bandwidth 192MHz) Standard: EN300 429/ITU-T J.83A/B MER: $\geq 40\text{db}$ RF frequency: 50~960MHz, 1KHz step RF output level: -20~+10dbm, 0.1db step Symbol Rate: 5.0M sps~7.0M sps, 1ksps stepping	QAM Channel: 16 non-adjacent carriers output (maximum bandwidth 192MHz) Standard: EN300 429/ITU-T J.83A/B MER: $\geq 40\text{db}$ RF frequency: 50~960MHz, 1KHz step RF output level: -20~+10dbm, 0.1db step Symbol Rate: 5.0M sps~7.0M sps, 1ksps stepping	QAM Channel: 16 non-adjacent carriers output (maximum bandwidth 192MHz) Standard: EN300 429/ITU-T J.83A/B MER: $\geq 40\text{db}$ RF frequency: 50~960MHz, 1KHz step RF output level: -20~+10dbm, 0.1db step Symbol Rate: 5.0M sps~7.0M sps, 1ksps stepping	QAM Channel: 16 non-adjacent carriers output (maximum bandwidth 192MHz) Standard: EN300 429/ITU-T J.83A/B MER: $\geq 40\text{db}$ RF frequency: 50~960MHz, 1KHz step RF output level: -20~+10dbm, 0.1db step Symbol Rate: 5.0M sps~7.0M sps, 1ksps stepping	QAM Channel: 16 non-adjacent carriers output (maximum bandwidth 192MHz) Standard: EN300 429/ITU-T J.83A/B MER: $\geq 40\text{db}$ RF frequency: 50~960MHz, 1KHz step RF output level: -20~+10dbm, 0.1db step Symbol Rate: 5.0M sps~7.0M sps, 1ksps stepping
Modulation	DVB-C				J.83A	J.83B
Modulation	DVB-C	Constellation	Constellation	Constellation	16/32/64/128/256QAM	64/256QAM
Modulation	DVB-C	Bandwidth	Bandwidth	Bandwidth	8M	6M
Modulation	DVB-T	Standard	Standard	Standard	EN300744	EN300744
Modulation	DVB-T	FFT mode	FFT mode	FFT mode	2K, 4K, 8K	2K, 4K, 8K
Modulation	DVB-T	Bandwidth	Bandwidth	Bandwidth	6M, 7M, 8M	6M, 7M, 8M
Modulation	DVB-T	Constellation	Constellation	Constellation	QPSK, 16QAM, 64QAM	QPSK, 16QAM, 64QAM
Modulation	DVB-T	Guard Interval	Guard Interval	Guard Interval	1/4, 1/8, 1/16, 1/32	1/4, 1/8, 1/16, 1/32

Modulation	DVB-T	FEC	FEC	FEC	1/2, 2/3, 3/4, 5/6, 7/8	1/2, 2/3, 3/4, 5/6, 7/8
Modulation	DVB-T	MER	MER	MER	≥42 dB	≥42 dB
Modulation	DVB-T	RF frequency	RF frequency	RF frequency	50~960MHz, 1KHz step	50~960MHz, 1KHz step
Modulation	DVB-T	RF out	RF out	RF out	8 non-adjacent carriers output (maximum bandwidth 192MHz)	8 non-adjacent carriers output (maximum bandwidth 192MHz)
Modulation	DVB-T	RF output level	RF output level	RF output level	-20~ +10 dBm, 0.1db step	-20~ +10 dBm, 0.1db step
Modulation	ATSC	Standard	Standard	Standard	ATSC A/53	ATSC A/53
Modulation	ATSC	Bandwidth	Bandwidth	Bandwidth	6M	6M
Modulation	ATSC	Constellation	Constellation	Constellation	8VSB	8VSB
Modulation	ATSC	FEC	FEC	FEC	RS(208 18 8)+Trellis	RS(208 18 8)+Trellis
Modulation	ATSC	MER	MER	MER	≥40dB	≥40dB
Modulation	ATSC	ACL	ACL	ACL	-55 dBc	-55 dBc
Modulation	ATSC	RF frequency	RF frequency	RF frequency	50~960MHz, 1KHz step	50~960MHz, 1KHz step
Modulation	ATSC	RF out	RF out	RF out	8 non-adjacent carriers output (maximum bandwidth 192MHz)	8 non-adjacent carriers output (maximum bandwidth 192MHz)
Modulation	ATSC	RF output level	RF output level	RF output level	-20~+10dbm (for all carriers), 0.5db stepping	-20~+10dbm (for all carriers), 0.5db stepping
Modulation	ISDB-T	Standard	Standard	Standard	ARIB STD-B31	ARIB STD-B31

Modulation	ISDB-T	Bandwidth	Bandwidth	Bandwidth	6M	6M
Modulation	ISDB-T	Constellation	Constellation	Constellation	QPSK, 16QAM, 64QAM	QPSK, 16QAM, 64QAM
Modulation	ISDB-T	Guard Interval	Guard Interval	Guard Interval	1/32, 1/16, 1/8, 1/4	1/32, 1/16, 1/8, 1/4
Modulation	ISDB-T	Transmission Mode	Transmission Mode	Transmission Mode	2K, 4K, 8K	2K, 4K, 8K
Modulation	ISDB-T	Code rate	Code rate	Code rate	1/2, 2/3, 3/4, 5/6, 7/8	1/2, 2/3, 3/4, 5/6, 7/8
Modulation	ISDB-T	MER	MER	MER	≥40dB	≥40dB
Modulation	ISDB-T	RF frequency	RF frequency	RF frequency	50~960MHz, 1KHz step	50~960MHz, 1KHz step
Modulation	ISDB-T	RF out	RF out	RF out	6 non-adjacent carriers output (maximum bandwidth 192MHz)	6 non-adjacent carriers output (maximum bandwidth 192MHz)
Modulation	ISDB-T	RF output level	RF output level	RF output level	-20dBm~+10dBm, 0.1dB stepping	-20dBm~+10dBm, 0.1dB stepping

Stream output	1 ASI output as mirror of one of RF output carriers-Optional 16 MPTS output over UDP and RTP/RTSP as mirror of 16 DVB-C carriers, 8 MPTS output over UDP and RTP/RTSP as mirror of 8 DVB-T/A TSC carriers(Optional) 6 MPTS output over UDP and RTP/RTSP as mirror of 6 ISDB-T carriers(Optional) 1*1000M Base-T Ethernet interface	1 ASI output as mirror of one of RF output carriers-Optional 16 MPTS output over UDP and RTP/RTSP as mirror of 16 DVB-C carriers, 8 MPTS output over UDP and RTP/RTSP as mirror of 8 DVB-T/A TSC carriers(Optional) 6 MPTS output over UDP and RTP/RTSP as mirror of 6 ISDB-T carriers(Optional) 1*1000M Base-T Ethernet interface	1 ASI output as mirror of one of RF output carriers-Optional 16 MPTS output over UDP and RTP/RTSP as mirror of 16 DVB-C carriers, 8 MPTS output over UDP and RTP/RTSP as mirror of 8 DVB-T/A TSC carriers(Optional) 6 MPTS output over UDP and RTP/RTSP as mirror of 6 ISDB-T carriers(Optional) 1*1000M Base-T Ethernet interface	1 ASI output as mirror of one of RF output carriers-Optional 16 MPTS output over UDP and RTP/RTSP as mirror of 16 DVB-C carriers, 8 MPTS output over UDP and RTP/RTSP as mirror of 8 DVB-T/A TSC carriers(Optional) 6 MPTS output over UDP and RTP/RTSP as mirror of 6 ISDB-T carriers(Optional) 1*1000M Base-T Ethernet interface	1 ASI output as mirror of one of RF output carriers-Optional 16 MPTS output over UDP and RTP/RTSP as mirror of 16 DVB-C carriers, 8 MPTS output over UDP and RTP/RTSP as mirror of 8 DVB-T/A TSC carriers(Optional) 6 MPTS output over UDP and RTP/RTSP as mirror of 6 ISDB-T carriers(Optional) 1*1000M Base-T Ethernet interface	1 ASI output as mirror of one of RF output carriers-Optional 16 MPTS output over UDP and RTP/RTSP as mirror of 16 DVB-C carriers, 8 MPTS output over UDP and RTP/RTSP as mirror of 8 DVB-T/A TSC carriers(Optional) 6 MPTS output over UDP and RTP/RTSP as mirror of 6 ISDB-T carriers(Optional) 1*1000M Base-T Ethernet interface
System function	Network management (WEB)	Network management (WEB)	Network management (WEB)	Network management (WEB)	Network management (WEB)	Network management (WEB)
System function	Chinese and English language	Chinese and English language	Chinese and English language	Chinese and English language	Chinese and English language	Chinese and English language
System function	Ethernet software upgrade	Ethernet software upgrade	Ethernet software upgrade	Ethernet software upgrade	Ethernet software upgrade	Ethernet software upgrade
Miscellaneous	Dimension (W×L×H)	Dimension (W×L×H)	482mm×440mm×44mm	482mm×440mm×44mm	482mm×440mm×44mm	482mm×440mm×44mm

Miscellaneous	Environment	Environment	0~45■(work)■-20~80 ■Storage ■	0~45■(work)■-20~80 ■Storage ■	0~45■(work)■-20~80 ■Storage ■	0~45■(work)■-20~80 ■Storage ■
Miscellaneous	Power requirements	Power requirements	AC 110V±10%, 50/60Hz, AC 220 ±10%, 50/60Hz	AC 110V±10%, 50/60Hz, AC 220 ±10%, 50/60Hz	AC 110V±10%, 50/60Hz, AC 220 ±10%, 50/60Hz	AC 110V±10%, 50/60Hz, AC 220 ±10%, 50/60Hz

Ordering Information

	IP input	IP output	RF carriers	Scrambling
DVB-C	512	16	16	√
DVB-T	256	8	8	×
ATSC	256	8	8	×
ISDB-T	192	6	6	×

About Us

FMUSER is a company specializing in broadcasting and television transmission systems and integrating hotel IPTV systems. The company was established in 2009 and has served over 180 countries worldwide. More than 20,000 television stations, radio stations, and hotel clients from different.

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